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CooPer plots – the next level of UpSet plots

Background

To visualize intersects between datasets, two main types of plots are available: Venn diagrams and UpSet plots. Although Venn diagrams provide a common solution to visualize the relation between 2-3 datasets, application in the context of $\geq$5 sets is usually not recommended. By contrast, UpSet plots are able to clearly visualize intersects between $\geq$5 datasets. However, the more sets and intersects are present, the more difficult it gets to visualize all information in a single plot. Furthermore, information on a second time point cannot be added. To overcome this current limitation, we developed CooPer plots (CO-Occurrence and PERsistence plots).

Methods

CooPer plots are based on the idea of interaction networks: We visualize every set as a node and every intersect between two sets as an edge. To illustrate the size of a set, it is correlated with the size of the node. Additionally, the exact numbers are printed within or next to the nodes. To illustrate the size of a specific intersect, the edges' thickness corresponds to the size of the intersects. If two sets do not share any intersect, they are not connected. Thereby, the complexity of the network is reduced.

In addition to the basic network, visualizing co-occurrence/intersects between sets, color coding allows us to add information on a second time point to the plot. If a user provides such information, the numbers characterizing the sets at the second time point are added to the nodes in parentheses. If a user opts for color coding, the nodes are colored from yellow (low number of persisting features) to dark red (high number of persisting features). The user may choose between coloring according to the relative or the absolute numbers. In addition, the edges are also color coded. Intersects at the first time point are visualized by gray edges, intersects at the second time point by black edges.

CooPer plots are generated with the help of the R package 'igraph' [1]. We developed a shiny app, freely available at https://sand-imi-uni-muenster.shinyapps.io/CooPerPlots/, to generate basic CooPer plots. In addition to the actual CooPer plot, a table with information on the size of each category at the first and second time point and the percentage of persisting elements is reported. To illustrate the use of CooPer plots, we provide a demo.

Results and Discussion

To show applicability of CooPer plots, we selected two exemplary datasets. The first set covers a large set of 537 MDS cases. For each patient, targeted sequencing information is publicly available at the SRA (BioProjectID: 388411). Additionally, variant calling results have been published [3]. Altogether, mutations in 19 genes are reported. Patients are characterized by zero to six mutations (average 1.58). Different from Venn diagrams or UpsetPlots the CooPer plot allows us to identify the top mutated genes (SF3B1, TET2 and ASXL1), the main connections (SF3B1 with ASXL1, and SF3B1 with TET2) and lacking co-occurrence (e.g. IDH1 and IDH2) quickly (see Figure 1).

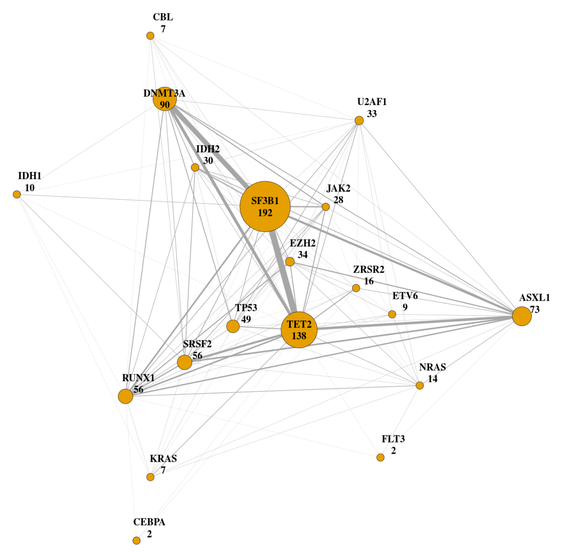

Figure 1: Visualization of co-occurring mutations in 537 cases with MDS.

The second set covers 117 cases with COVID-19 [2]. For each patient, information on initial and persisting symptoms was analyzed. Altogether, 17 different symptoms were mentioned in the study. Patients are characterized by zero to seven symptoms at time point 1 (average 3.21) and zero to two symptoms at time point 2 (average 0.25).

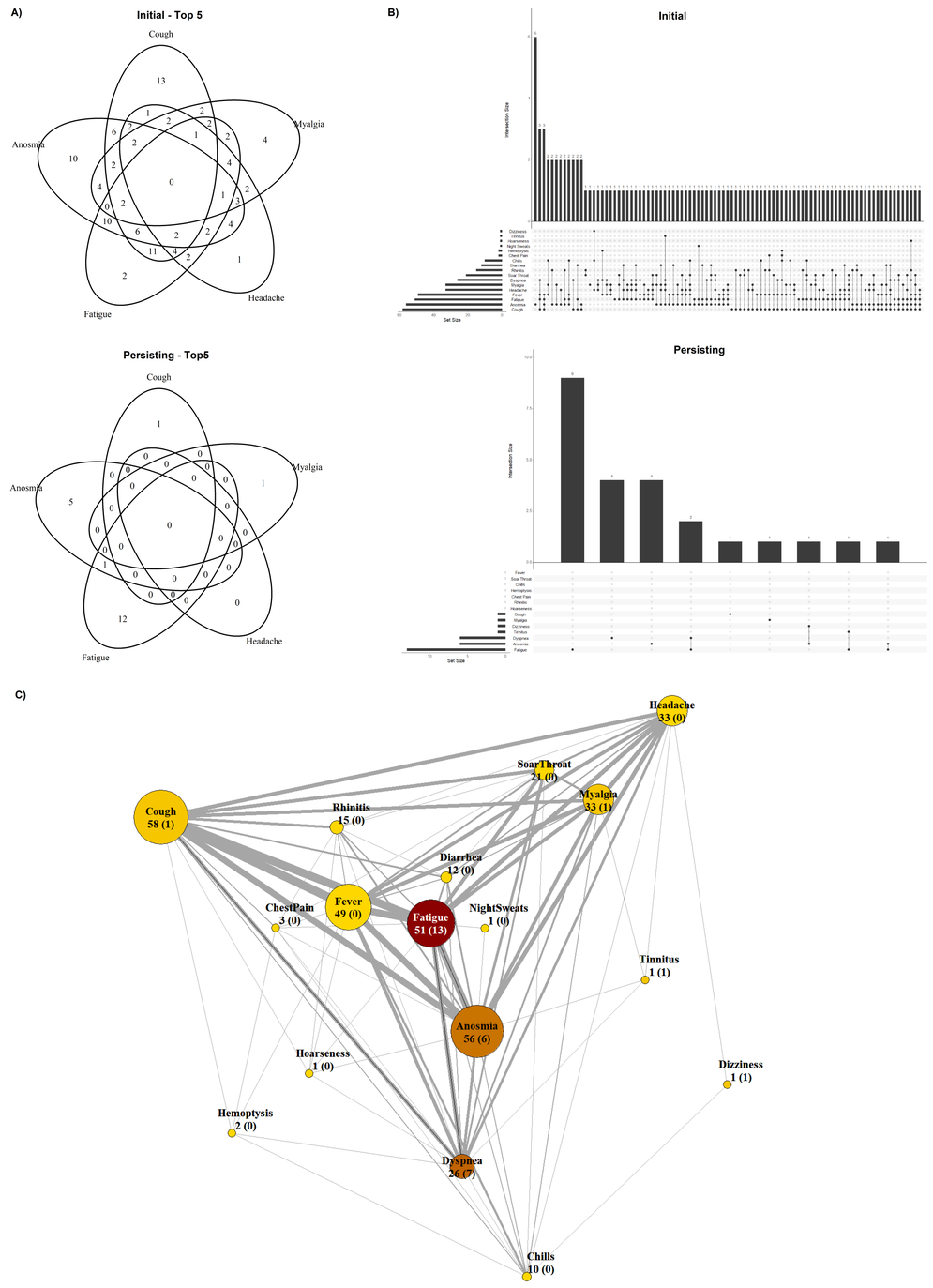

Figure 2: Visualization of initial vs presisting symptoms in 117 cases with COVID-19. A) Venn diagrams showing the top 5 symptoms, B) UpSet plots showing all symptoms and intersects, C) CooPer plot.

For the given use-case, considering two time points, CooPer plots provide an improved way of displaying data: At once, it can be observed that cough, anosmia, fatigue and fever are the main symptoms (see Figure 2). Especially anosmia and fatigue are characterized as persisting symptoms. All main symptoms regularly co-occur. However, e.g. dizziness and anosmia never co-occur.

Conclusion

CooPer plots extend the general idea of interaction networks, displaying scientific data with special focus on co-occurring and persisting features. Applicability covers a diverse set of data. Two use-cases show clear superiority of CooPer plots over Venn diagrams. Furthermore, advantages in comparison to UpSet plots can be observed in the presence of a second time point or data characterized by many sets and intersects.

References

[1] Csardi, G. and Nepusz, T. (2006) The igraph software package for complex network research. *InterJournal*, **Complex Systems**, 1695; https://igraph.org. **[2]** Varghese, J. *et al.* (2021) Persistent symptoms and lab abnormalities in patients who recovered from COVID-19. *Sci Rep*, **11**, 12775; doi: 10.1038/s41598-021-91270-8. **[3]** Sandmann, S. *et al.* (2018) appreci8: a pipeline for precise variant calling integrating 8 tools. *Bioinformatics*, **34**, 4205-4212; doi: 10.1093/bioinformatics/bty518.

Availability: The CooPer plots shiny app is freely available at https://sand-imi-uni-muenster.shinyapps.io/CooPerPlots/. CooPer plots is for research use only.

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